

Sasun Madhuranga

SOFTWARE ENGINEER | FULL-STACK WEB DEVELOPER

▪ Kandy, Sri Lanka ▪ sasunmadhuranga@gmail.com ▪ +94775789203

▪ LinkedIn: [linkedin.com/in/sasun-madhuranga](https://www.linkedin.com/in/sasun-madhuranga) ▪ GitHub: github.com/sasunmadhuranga

▪ Portfolio: <https://sasunmadhuranga.vercel.app>

Full-Stack Developer with a BSc in Information Technology and experience building scalable web and mobile applications using the MERN stack. Developed RESTful APIs, microservices architecture, and JWT-based authentication systems, with production deployments on AWS and Microsoft Azure using Docker and GitHub Actions CI/CD. Delivered systems supporting 1,000+ concurrent users across AI chatbot, legal tech, and volunteer management platforms.

Technical Skills

- **Languages:** JavaScript, TypeScript, Python, Java, PHP, SQL, HTML, CSS
 - **Frontend:** React.js, Next.js, Tailwind CSS
 - **Backend:** Node.js, Express.js, REST APIs, FastAPI
 - **Databases:** MongoDB, MySQL
 - **Cloud & DevOps:** AWS, Microsoft Azure, Docker, CI/CD (GitHub Actions), Linux, Terraform
 - **Authentication & Security:** JWT, Role-Based Access Control (RBAC), API Security
 - **Concepts:** Microservices Architecture, MVC, OOP, Version Control (Git)
 - **Other:** Artificial Intelligence (AI), Generative AI, React Native, Rasa
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Professional Experience

AI/ML Intern | ARTecX Solutions

Sept 2025 – Mar 2026

- Designed AI-powered chatbot systems using Python and FastAPI, automating 100% of user query handling.
 - Built and containerized microservices using Docker and deployed cloud-native applications on Microsoft Azure.
 - Implemented real-time location tracking using geospatial data and interactive maps, supporting 1,000+ concurrent requests.
 - Collaborated with cross-functional teams following Agile methodology and SDLC practices to deliver scalable software solutions.
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Projects & Technical Experience

Volunteer Management Platform

Sept 2025 – Jan 2026

MongoDB, Express.js, React.js, Node.js, React Native, Tailwind CSS, JWT, QR

- Built a full-stack MERN-based volunteer platform with a React Native mobile app.
- Developed and incorporated 9+ core features, including event management, volunteer onboarding, attendance tracking, and certificate generation, improving overall system efficiency.
- Enforced role-based access control, enabling 100% secure, approval-based event workflows and reducing unauthorized access incidents to zero.
- Constructed a secure QR-based attendance system using daily, time-bound tokens, reducing manual attendance errors by 70%+.
- Launched the system on Vercel, Render, MongoDB Atlas, and Cloudinary, reducing load times by 40% and enabling reliable cross-platform synchronization for web and mobile users.

Instagram Clone

May 2025 – Aug 2025

MongoDB, Express.js, React.js, Node.js, Tailwind CSS, JWT, Docker, AWS EKS, S3, Terraform, Jenkins

- Engineered a fully responsive Instagram clone, with features for media upload, post creation, and 99.9% system uptime.
- Implemented JWT-based authentication and protected RESTful API routes to ensure secure user sessions and data integrity.
- Replicated over 6 core features such as user posts, likes, comments, and profiles, and related sub-features.
- Containerized the Node.js/Express backend with Docker, hosting backend services on Amazon EKS and the React frontend on Amazon S3.
- Provisioned AWS infrastructure using Terraform and automated deployments via a Jenkins CI/CD pipeline connected to GitHub webhooks.

Legal Assistance App with AI chatbot

Aug 2024 – Aug 2025

React.js, Python, PHP, HTML/CSS, Rasa, JavaScript, Axios

- Created a legal support platform for consultation, document creation, and case handling.
- Integrated an AI-powered chatbot using the Rasa framework, enabling 24/7 automated legal support.
- Achieved a 65%+ resolution rate for user queries and reduced human support workload by 40% and average query response time.

Education

BSc in Information Technology (Final Year GPA: 3.64)

2021 - 2025

University of Moratuwa, Sri Lanka